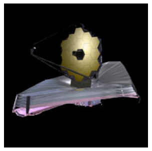

# 9492 - From Rags to Calcium-riches: Using JWST to Constrain Progenitor Identity and Dust Formation in the Calcium-strong Transient 2025coe

Cycle: 4, Proposal Category: DD

**INVESTIGATORS**

| ***Name*** | ***Institution*** |
|---|---|
| **Dr. Wynn Vicente Jacobson-Galan (PI)** | **California Institute of Technology** |
| Prof. Alex V. Filippenko (CoI) | University of California - Berkeley |
| Dr. Luc Dessart (CoI) (ESA Member) | CNRS, Institut d'Astrophysique de Paris |
| Prof. Ryan Foley (CoI) | University of California - Santa Cruz |
| Dr. Andy Howell (CoI) | Las Cumbres Observatory Global Telescope Network |
| Dr. Saurabh W. Jha (CoI) | Rutgers the State University of New Jersey |
| Conor Larison (CoI) | Space Telescope Science Institute |
| Mr. Chang Liu (CoI) | Northwestern University |
| Dr. Adam Miller (CoI) | Northwestern University |
| Prof. Maryam Modjaz (CoI) | The University of Virginia |
| Dr. Craig Pellegrino (CoI) | The University of Virginia |
| Prof. David J. Sand (CoI) | University of Arizona |
| Dr. Huei Sears (CoI) | Rutgers the State University of New Jersey |
| Prof. Stefano Valenti (CoI) | University of California - Davis |
| Dr. Steve Schulze (CoI) | Northwestern University |
| Prof. Griffin Hosseinzadeh (CoI) | University of California - San Diego |
| Prof. Daichi Hiramatsu (CoI) | University of Florida |
| Dr. Mansi Kasliwal (CoI) | California Institute of Technology |
| Ferdinand Ferdinand (CoI) | University of Illinois at Urbana - Champaign |
| Oleksandra Pyshna (CoI) | California Institute of Technology |
| Rujula Yete (CoI) | University of North Carolina at Chapel Hill |

| *Name* | *Institution* |
|---|---|
| Ms. Saarah Hall (CoI) | Northwestern University |
| Ms. Sam Rose (CoI) | California Institute of Technology |
| Dr. Lindsey Kwok (CoI) | Northwestern University |
| Sahana Kumar (CoI) | The University of Virginia |


**OBSERVATIONS**

| *Folder* | *Observation* | *Label* | *Observing Template* | *Science Target* |
|---|---|---|---|---|
| Late-time NIRSpec PRISM + MIRI LRS | | | | |
| | 1 | NIRSpec PRISM | NIRSpec Fixed Slit Spectroscopy | (1) SN2025COE |
| | 2 | MIRI LRS | MIRI Low Resolution Spectroscopy | (1) SN2025COE |

## ABSTRACT

Supernova (SN) 2025coe is a recently identified Calcium-strong transient (CaST) located at a distance of ~26 Mpc, making it the second closest CaST discovered to date. Despite of its extremely remote location at ~40 kpc offset from host galaxy NGC 3277, SN 2025coe's double-peaked multi-band light curve and luminous early-time X-ray luminosity (only the third ever X-ray detection to date) suggests a progenitor system involving dense circumstellar material (CSM), similar to that of other CSM-interacting CaSTs (e.g., iPTF16hgs, 2019ehk, 2021gno, 2022oqm). Given its proximity, SN 2025coe represents an unprecendented opportunity for JWST observations in order to obtain the (1) latest NIR spectrum of a CaST to date, and (2) the first ever MIR spectrum of a CaST. We propose 6 hours of DD time to observe SN 2025coe with JWST NIRSpec + MIRI LRS in order to detect forbidden line emission in the NIR+MIR in order to robustly constrain the CaST progenitor system through state-of-the-art non-LTE radiative transfer modeling. Furthermore, SN 2025coe represents a unique opportunity to search for emission from newly formed dust, a phenomenon that has only recently been confirmed through NIR spectroscopy of the CaST 2024uj. Given that CaSTs this nearby only occur every ~6 years, it is imperative that novel observations of SN 2025coe be conducted with JWST in order to progress our understanding of this peculiar explosion class.

## OBSERVING DESCRIPTION

We will observe the nearby Ca-strong transient SN 2025coe at one late time epoch (approximately 272+/-7 days post-discovery) with MIRI LRS and NIRSpec PRISM.

The late-time visit will use the NIRSpec PRISM and MIRI LRS to obtain spectroscopic coverage from 0.6 to 13 microns at low resolution (R ~ 100). The NIRSpec observations will cover warm thermal dust emission as well as the CO fundamental band and first overtone emission. The MIRI observations will also cover warm dust but additionally cover emission from other molecules such as SiO, SO, SiS, and PAHs. This late time observation will constrain dust and molecule growth in the SN ejecta. These observations will capture important SN emission lines that will allow us to distinguish between different models for the origins of Ca-strong transients.

Our observations will be non-disruptive.

**Fixed Targets**

| # | Name | Target Coordinates | Targ. Coord. Corrections | Miscellaneous |
|---|---|---|---|---|
| (1) | SN2025COE | RA: 10 33 7.9237 (158.2830154d)<br>Dec: +28 26 12.60 (28.43683d)<br>Equinox: J2000 | Epoch of Position: 2000 | |

*Comments:*
*Category=Star*
*Description=[Supernovae, Type Ia supernovae, Type Ib supernovae]*
*Extended=NO*

## Observation

**Proposal 9492, Observation 1: NIRSpec PRISM**

Tue Oct 21 14:00:18 GMT 2025

**Diagnostic Status: Warning**

Observing Template: NIRSpec Fixed Slit Spectroscopy

## Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 1:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.

## Fixed Targets

| # | Name | Target Coordinates | Targ. Coord. Corrections | Miscellaneous |
|---|---|---|---|---|
| (1) | SN2025COE | RA: 10 33 7.9237 (158.2830154d)<br>Dec: +28 26 12.60 (28.43683d)<br>Equinox: J2000 | Epoch of Position: 2000 | |

*Comments:*
*Category=Star*
*Description=[Supernovae, Type Ia supernovae, Type Ib supernovae]*
*Extended=NO*

## Acquisition

| # | Target | TA Method | Subarray | Filter | Readout Pattern | Groups/Int | Integrations/Exp | Total Integrations | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | SAME | WATA | FULL | CLEAR | NRSRAPIDD6 | 3 | 1 | 1 | 171.788 | 263196 |

## Template

| HFF Readout Mode | Slit | Subarray |
|---|---|---|
| false | S200A1 | FULL |

## Dithers

| # | Primary Dither Positions | Sub-Pixel Pattern |
|---|---|---|
| 1 | 3 | NONE |

## Spectral Elements

| # | Grating/Filter | Slit | Readout Pattern | Groups/Int | Integrations/Exp | # | Autocal | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | PRISM/CLEAR | S200A1 | NRSIRS2RAPID | 50 | 1 | 1 | NONE | 3 | 3 | 2232.1 | 263196 |

| Special Requirements | |
| --- | --- |
| | Between Dates 01-NOV-2025:00:00:00 and 30-NOV-2025:00:00:00 |

## Observation

**Proposal 9492, Observation 2: MIRI LRS**

Tue Oct 21 14:00:18 GMT 2025

**Diagnostic Status: Warning**

Observing Template: MIRI Low Resolution Spectroscopy

## Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 2:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.

## Fixed Targets

| # | Name | Target Coordinates | Targ. Coord. Corrections | Miscellaneous |
|---|---|---|---|---|
| (1) | SN2025COE | RA: 10 33 7.9237 (158.2830154d)<br>Dec: +28 26 12.60 (28.43683d)<br>Equinox: J2000 | Epoch of Position: 2000 | |

*Comments:*
*Category=Star*
*Description=[Supernovae, Type Ia supernovae, Type Ib supernovae]*
*Extended=NO*

## Acquisition

| # | Target | Filter | Readout Pattern | Groups/Int | Integrations/Exp | Total Integrations | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|
| 1 | SAME | F1000W | FASTGRPAVG32 | 10 | 1 | 1 | 888.013 | 263196 |

## Template

| Subarray | Obtain Verification Image? |
|---|---|
| FULL | true |

## Dithers

| # | Dither Type | No. Spectral Steps | Spectral Step Offset | No. Spatial Steps | Spatial Step Offset |
|---|---|---|---|---|---|
| 1 | ALONG SLIT NOD | | | | |

## Pointing Verification

| # | PV Readout Pattern | PV Groups/Int | PV Integrations/Exp | PV Total Integrations | PV Exposures/Dith | PV Total Dithers | PV Total Exposure Time | Optional ETC ID | Filter |
|---|---|---|---|---|---|---|---|---|---|
| 1 | FASTR1 | 128 | 2 | 2 | 1 | 1 | 713.185 | | F1000W |

**Spectral Elements**

| # | Readout Pattern | Groups/Int | Integrations/Exp | Total Integrations | Exposures/Dith | Total Dithers | Total Exposure Time | Optional ETC ID |
|---|---|---|---|---|---|---|---|---|
| 1 | FASTR1 | 90 | 20 | 40 | 1 | 2 | 10095.596 | 263196 |

**Special Requirements**

Between Dates 01-NOV-2025 and 30-NOV-2025